

Alexander Joos

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EDUCATION

Cornell University

MEng Computer Science

Ithaca, NY

Jan. 2026 – Dec. 2026

Cornell University

B.A. Information Science (Data Science)

Ithaca, NY

Aug. 2022 – Dec. 2025

National University of Singapore

Yale NUS College Exchange Program (Computer Science)

Singapore

Jan. 2025 – May 2025

Relevant Coursework: Graduate Machine Learning, LLMs, Graduate Software Engineering, Data Science, Backend Development, Graduate Algorithm Design, Graduate Deep Learning, Statistics, Linear Algebra, Computer Organization

SKILLS

Languages: Python, Java, SQL, OCaml, C/C++, R, JavaScript, HTML/CSS

ML & AI: PyTorch, transformers, scikit-learn, LangChain, OpenAI & Anthropic APIs, MCP, Vector Database, RAG

Backend & Data: Flask, REST APIs, pandas, SQLAlchemy, BigQuery, Vertex AI, Looker Oracle, PostgreSQL, Snowflake

Developer Tools: Git, GitHub, Docker Compose, Apache Airflow, Kafka, Bash, Jupyter, Google Colab, Jira, Confluence

EXPERIENCE

Fidelity Investments

June 2026 – August 2026

Graduate Data Engineering Intern, Database Modernization

Merrimack, NH

- Developed low-latency, deterministic distributed brokerage platform & streaming state machine exhaust to SingleStore.
- Built streaming pipelines for capital markets data with Aeron and Kafka, & ETL pipelines from Kafka into SingleStore.
- Developed validation agents & tool-call workflows for agentic legacy-code modernization using TLA+ for design verification.
- Engineered high-throughput Kafka producer over Aeron with async fault-tolerant delivery for a sub-3ms brokerage platform.

Fidelity Investments

June 2025 – Aug. 2025

Data Engineering Intern, Cloud and Platform Engineering

Merrimack, NH

- Built and launched a Flask/AJAX web app on an Ubuntu VM to automate internal database model comparison workflows.
- Supported brokerage platform model analysis across Oracle, PostgreSQL, CockroachDB, & Snowflake databases.
- Designed Flask REST API endpoints & Python/SQL backend storage/retrieval behind JavaScript/HTML/CSS frontend.
- Deployed the application through Apache, SSL certificate setup, firewall whitelisting, and DEV/PROD rollout.

American Family Insurance: Homesite

June 2023 – Aug. 2024

Partner Distribution Intern → Data Analytics Intern

Boston, MA

- Built BigQuery pipelines from raw source tables for a Fortune 50 retailer's Partner Management reporting layer in SQL.
- Built 8 Looker dashboards for Partner & Executive teams; automated SQL query generation via Excel/Sheets scripts.
- Conducted Excel, Power Pivot, & Power Query analysis on top U.S. auto carrier to identify competition feature importance.

PROJECTS

Apache Airflow Anomaly Detection – Agile Open Source Systems & Backend Feature Development

May 2026

- Built Airflow-native task-runtime anomaly detection with DAG-file config, custom detectors, & 4 built-in algorithms.
- Implemented backend/task SDK, structured results, task-level observability, CI tests, & example DAGs/docs.

LOB-Driven Optimal Execution Strategy – Quantitative Trading & Market Microstructure

May 2026

- Designed TWAP-beating execution strategy using order-book imbalance & spread-threshold signals on tick-level LOB data.
- Backtested on INTC/MSFT/AMZN/GOOG with smoothed grid search and separate parameter fits by spread regime.

LLM Text Watermarking – ML Research Reimplementation

May 2026

- Reimplemented Kirchenbauer et al. LLM watermarking in PyTorch with soft green-list logit biasing and z-test detection.

Clustering Analysis on Workplace Gender & Power – NLP & Unsupervised ML

Dec. 2024

- Cleaned email bodies into a high-dimensional TF-IDF matrix, capturing sentiment across 15,000+ messages.
- Used DBSCAN, K-means, RiveterNLP, and cross-tabulation approaches to analyze gender, power, and positional clusters.

Old Bailey Verdict Prediction Model – LLM Fine-Tuning & Classification

Nov. 2024

- Fine-tuned a Hugging Face LLM on 300+ years of > 100,000 Old Bailey case briefs across 8 legal categories with 71% F1.